

MEA OCPP 1.6 Compliance Test Report

Section 3: Normal Operation Check

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

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1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Connection	Direct (no proxy)
Test PC	192.168.111.185 (alias 192.168.1.200/24 on enp3s0)
EV Plug Timeout	0 s per plug event
Local Stop Timeout	30 s (RFID card tap)
Test Date	2026-06-10 08:33:26 UTC
Results file	tex/vsecc_section3_results.json

Test Architecture

The test runs directly against the MEA CSMS with no proxy. Two charging sessions are tested:

- **Manual session (3.1–3.10):** EV plugged in; driver taps RFID card to authorize and start; driver taps card again to stop locally.
- **Remote session (3.11–3.19):** EV plugged in; CSMS sends RemoteStartTransaction via MEA REST API; CSMS sends RemoteStopTransaction to stop.

MQTT subscription to `mqtt://192.168.1.166:1883` is used to observe all StatusNotification, Authorize, StartTransaction, StopTransaction, and MeterValues events.

2 Section 3 Results

Summary: **3 PASS** **0 FAIL** **16 WARN** **0 SKIP** (19 total)

Item	Test Description	Detail / Evidence	Result
Manual Operation Flow (3.1–3.10)			
3.1	BootNotification → Accepted (currentTime, interval, status)	CSMS connection active BootNotification Accepted confirmed	PASS
3.2	StatusNotification Available (initial state)	Available not observed on MQTT in 10 s <i>Check CSMS event log for StatusNotification Available on boot</i>	WARN
3.3	StatusNotification Preparing (EV plug, manual)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
3.4	Authorize (RFID card) → Accepted	Depends on 3.3 (EV plug)	WARN
3.5	StartTransaction (manual, meterStart, idTag, connectorId)	Depends on 3.3 (EV plug)	WARN
3.6	StatusNotification Charging (manual session)	Depends on 3.3 (EV plug)	WARN
3.7	MeterValues (energy measurands, manual session)	Depends on 3.3 (EV plug)	WARN
3.8	StopTransaction (manual card tap, reason=Local)	Depends on 3.3 (EV plug)	WARN
3.9	StatusNotification Finishing (manual, EV still plugged)	Depends on 3.3 (EV plug)	WARN
3.10	StatusNotification Available (manual, EV unplug)	Depends on 3.3 (EV plug)	WARN
Remote Operation Flow (3.11–3.19)			

Item	Test Description	Detail / Evidence	Result
3.11	StatusNotification Preparing (EV plug, remote)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
3.12	RemoteStartTransaction → Accepted	HTTP 200	PASS
3.13	StartTransaction (triggered by RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
3.14	StatusNotification Charging (remote session)	Charging status not observed on MQTT in 20 s	WARN
3.15	MeterValues (energy measurands, remote session)	MeterValues not observed on MQTT in 15 s	WARN
3.16	RemoteStopTransaction → Accepted	HTTP 200	PASS
3.17	StopTransaction (triggered by RemoteStop)	session_state idle/finished not observed on MQTT in 25 s	WARN
3.18	StatusNotification Finishing (remote, EV still plugged)	Finishing not observed on MQTT in 15 s <i>Some implementations skip Finishing → go directly to Available</i>	WARN
3.19	StatusNotification Available (remote, EV unplug)	Available not observed on MQTT in 30 s unplug EV	WARN

3 Notes on Failures and Warnings

EV-dependent items

- **3.3, 3.11** StatusNotification Preparing: require a physical EV to plug into the vSECC connector within the EV_WAIT_SEC timeout.
- **3.4–3.10, 3.13–3.19** All session items depend on the EV being connected in items 3.3 / 3.11 respectively.

Item 3.8 — Manual stop

Item 3.8 tests that the EV driver can tap their RFID card on the vSECC to trigger a local **StopTransaction** with **reason=Local**. If no card tap is detected within **LOCAL_STOP_WAIT_SEC** seconds, the test falls back to **RemoteStop** to clear the session so the remote-session phase can proceed. The fallback is recorded as **WARN** with a remark.

Items 3.7, 3.15 — MeterValues

The MEA sandbox REST API returns HTTP 404 for **TriggerMessage**. **MeterValues** are observable only when the vSECC autonomously publishes them to MQTT based on the **MeterValueSampleInterval** configuration (set in Section 1 item 1.11).

Items 3.12, 3.16 — RemoteStart / RemoteStop

- **3.12** Uses a dedicated password (**MEA_PASS_START**) for the **/EV/cmd/chargepoint/remoteStart** endpoint.
- **3.16** **RemoteStop** with **transaction_id=0** succeeds when exactly one transaction is active; **FAIL** if no active transaction exists.

4 Dependency Map

Manual 3.3 → 3.4 → 3.5 → 3.6 → 3.7 → 3.8 → 3.9 → 3.10

Remote 3.11 → 3.12 → 3.13 → 3.14 → 3.15 → 3.16 → 3.17 → 3.18 → 3.19

Items 3.1 and 3.2 are independent of EV presence.